Nmap Scanning and Network Traffic Analysis

Topic: Network Traffic Analysis and Nmap.
Task: Perform Nmap scans against Ubuntu and Windows virtual machines

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1. Objective

The objective of this practical was to use Nmap from Kali Linux to discover hosts, identify open ports, detect running services, and estimate the operating systems of the Ubuntu and Windows virtual machines.

The practical also involved capturing a TCP SYN scan using Wireshark to observe how Nmap scan traffic appears at the packet level.

The main objectives were:

- Perform host discovery using -sn.
- Perform a TCP SYN scan using -sS.
- Identify services and versions using -sV.
- Perform operating system detection using -O.
- Perform aggressive detection using -A.
- Save scan results in normal text and XML formats.
- Capture and analyze TCP SYN scan packets in Wireshark.

2. Lab Environment

Component	Description
Kali Linux	Scanning machine
Ubuntu Linux	Target machine 1
Windows	Target machine 2
Nmap	Network discovery and scanning tool
Wireshark	Packet capture and analysis tool

Network configuration

- Kali Linux IP: [Enter Kali IP]
- Ubuntu IP: [Enter Ubuntu IP]
- Windows IP: [Enter Windows IP]
- Network interface: NAT

All scans were intended for the authorized virtual machines in the lab environment.

3. Preparation

Nmap was installed on Kali Linux using:

```
sudo apt update
sudo apt install nmap -y
```

The installed version was verified with:

```
nmap --version
```

The IP addresses of the Ubuntu and Windows virtual machines were identified before scanning.

4. Nmap Scans Performed

4.1. Host Discovery -sn

Command:

sudo nmap -sn -oN ubuntu-sn.txt -oX ubuntu-sn.xml 192.168.195.133
sudo nmap -sn -oN windows -sn.txt -oX windows-sn.xml 192.168.195.134

Purpose:

The -sn option performs host discovery without performing a port scan. Depending on the network configuration, Nmap may use ARP, ICMP, or TCP-based probes to determine whether a host is responding.

Observation:

The output was examined to determine whether Nmap considered the target host up.

Output files:

- ubuntu-sn.txt
- ubuntu-sn.xml
- windows-sn.txt
- windows-sn.xml

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sn 192.168.195.133 -oN ubuntu-sn.txt -oX ubuntu-sn.xml
[sudo] password for shivaay:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:51 +0530
Nmap scan report for 192.168.195.133
Host is up (0.0030s latency).
MAC Address: 00:0C:29:E3:49:EE (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds
```

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sn 192.168.195.134 -oN windows-sn.txt -oX windows-sn.xml
[sudo] password for shivaay:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 13:38 +0530
Nmap scan report for 192.168.195.134
Host is up (0.0073s latency).
MAC Address: 00:0C:29:CC:7D:6D (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.81 seconds
```

4.2.TCP SYN Scan -sS

Command:

sudo nmap -sS -oN ubuntu-sS.txt -oX ubuntu-sS.xml 192.168.195.133
sudo nmap -sS -oN windows -sS.txt -oX windows -sS.xml 192.168.195.134

Purpose:

The SYN scan sends TCP SYN packets to the selected ports and analyzes the responses to determine their likely state. It normally does not complete the full TCP handshake on open ports.

Typical responses include:

- **SYN-ACK:** The port is likely open.
- **RST:** The port is likely closed.
- **No response or certain ICMP errors:** The port may be filtered or otherwise difficult to classify.

Observation:
The scan results were examined to identify open, closed, or filtered ports.

Output files:
- ubuntu-sS.txt
- ubuntu-sS.xml
- windows-sS.txt
- windows-sS.xml

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sS 192.168.195.133 -oN ubuntu-sS.txt -oX ubuntu-sS.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:52 +0530
Nmap scan report for 192.168.195.133
Host is up (0.0026s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 00:0C:29:E3:49:EE (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds
```

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sS 192.168.195.134 -oN windows-sS.txt -oX windows-sS.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 13:38 +0530
Nmap scan report for 192.168.195.134
Host is up (0.0017s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
3389/tcp open  ms-wbt-server
5357/tcp open  wsdapi
MAC Address: 00:0C:29:CC:7D:6D (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.51 seconds
```

4.3.Service and Version Detection -sV

Command:

sudo nmap -sV -oN ubuntu-sV.txt -oX ubuntu-sV.xml 192.168.195.133
sudo nmap -sV -oN windows -sV.txt -oX windows -sV.xml 192.168.195.134

Purpose:
The -sV option probes discovered open ports to identify the services running on them and, when possible, their versions.

For example, it may identify an HTTP service on TCP port 80 or an SSH service on TCP port 22.

Observation:
The reported service names, versions, and port states were recorded.

Output files:

- ubuntu-sV.txt
- ubuntu-sV.xml
- windows-sV.txt
- windows-sV.xml

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sV 192.168.195.133 -oN ubuntu-sV.txt -oX ubuntu-sV.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:53 +0530
Nmap scan report for 192.168.195.133
Host is up (0.00029s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.66 ((Ubuntu))
MAC Address: 00:0C:29:E3:49:EE (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.48 seconds
```

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -sV 192.168.195.134 -oN windows-sV.txt -oX windows-sV.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 13:39 +0530
Nmap scan report for 192.168.195.134
Host is up (0.00076s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
3389/tcp open  ms-wbt-server Microsoft Terminal Services
5357/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 00:0C:29:CC:7D:6D (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.80 seconds
```

4.4.Operating System Detection -O

Command:

sudo nmap -O -oN ubuntu-O.txt -oX ubuntu-O.xml 192.168.195.133
sudo nmap -O -oN windows -O.txt -oX windows -O.xml 192.168.195.134

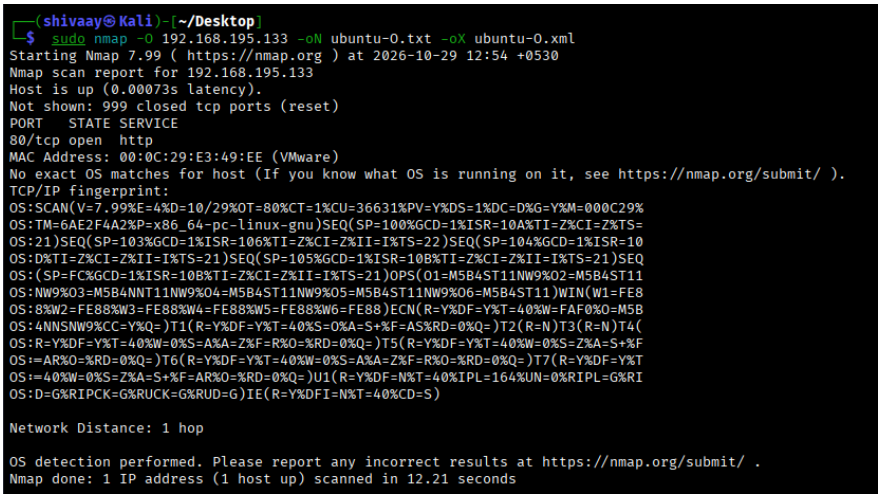
Purpose:
The -O option uses network-response characteristics to estimate the target operating system. Detection depends on available responses and suitable open and closed ports.

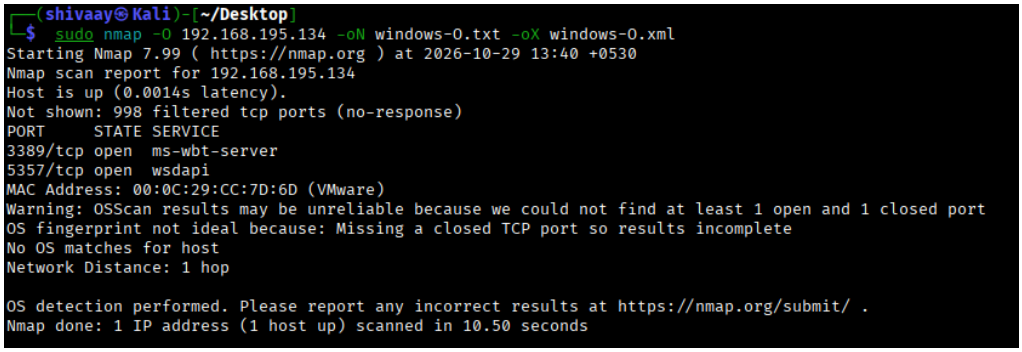
Observation:
The estimated operating system and any confidence information reported by Nmap were reviewed.

Output files:

- ubuntu-O.txt
- ubuntu-O.xml

- windows-O.txt
- windows-O.xml





An operating system estimate is not guaranteed to be correct. Virtualization, firewalls, and network configurations can affect the result.

4.5. Aggressive Detection - -A

Command:

sudo nmap -A -oN ubuntu-A.txt -oX ubuntu-A.xml 192.168.195.133

sudo nmap -A -oN windows -A.txt -oX windows -A.xml 192.168.195.133

Purpose:

The -A option enables several detection features, including operating system detection, service/version detection, default NSE scripts, and traceroute where supported.

Observation:

The service information, operating system estimate, script output, and traceroute information were reviewed where available.

Output files:

- ubuntu-A.txt
- ubuntu-A.xml

- windows-A.txt
- windows-A.xml

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -A 192.168.195.133 -oN ubuntu-A.txt -oX ubuntu-A.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 12:55 +0530
Nmap scan report for 192.168.195.133
Host is up (0.0014s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.66 ((Ubuntu))
|_http-server-header: Apache/2.4.66 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 00:0C:29:E3:49:EE (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=10/29%OT=80%CT=1%CU=35925%PV=Y%DS=1%DC=D%G=Y%M=000C29%
OS:TM=6AE2F4E6%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10C%TI=Z%CI=Z%TS=
OS:21)SEQ(SP=100%GCD=1%ISR=10F%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=103%GCD=1%ISR=10
OS:E%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=105%GCD=1%ISR=10A%TI=Z%CI=Z%TS=21)SEQ(SP=F
OS:3%GCD=1%ISR=109%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW9%O2=M5B4ST11NW9%O
OS:3=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WIN(W1=FE88%W2=
OS:FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSN
OS:W9%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%D
OS:F=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O
OS:=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W
OS:=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%R
OS:IPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   1.43 ms 192.168.195.133

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.05 seconds
```

```
┌──(shivaay㉿Kali)-[~/Desktop]
└─$ sudo nmap -A 192.168.195.134 -oN windows-A.txt -oX windows-A.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 13:41 +0530
Nmap scan report for 192.168.195.134
Host is up (0.0014s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
3389/tcp open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: DESKTOP-7BFC9O7
|   NetBIOS_Domain_Name: DESKTOP-7BFC9O7
|   NetBIOS_Computer_Name: DESKTOP-7BFC9O7
|   DNS_Domain_Name: DESKTOP-7BFC9O7
|   DNS_Computer_Name: DESKTOP-7BFC9O7
|   Product_Version: 10.0.19041
|_  System_Time: 2026-10-29T08:12:05+00:00
| ssl-cert: Subject: commonName=DESKTOP-7BFC9O7
| Not valid before: 2026-06-08T15:06:31
|_Not valid after:  2026-12-08T15:06:31
|_ssl-date: 2026-10-29T08:12:09+00:00; 0s from scanner time.
5357/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Service Unavailable
MAC Address: 00:0C:29:CC:7D:6D (VMware)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

TRACEROUTE
HOP RTT     ADDRESS
1   1.43 ms 192.168.195.134

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.74 seconds
```

Aggressive detection generates more probes than a basic scan and may be more noticeable in network monitoring logs.

5. Wireshark Capture of the SYN Scan

Procedure

I. Wireshark was opened on Kali Linux.
II. The interface connected to the target VM network was selected.
III. Packet capture was started.
IV. A TCP SYN scan was launched against the Ubuntu VM:

i. sudo nmap -sS -p 1-1000 192.168.195.133

V. The capture was stopped after the scan completed.

VI. The following display filter was applied:

i. tcp.flags.syn == 1 && tcp.flags.ack == 0

VII. The relevant packet sequence was inspected, and the capture was saved as ubuntu-sS-scan.pcap

Packet Analysis

The display filter selects TCP packets with the SYN flag set and the ACK flag unset. These packets commonly represent initial TCP connection attempts.

During a SYN scan, the following behavior may be observed:

Packet or response	Interpretation
SYN from Kali	Scanner initiates a TCP probe
SYN-ACK from target	Port is likely open
RST from target	Port is likely closed
No response	Port may be filtered or the probe may have been dropped
RST from scanner after SYN-ACK	Scanner terminates the half-open connection attempt

Not every port produces all these packets. Firewalls, packet loss, and target behavior can affect the observed sequence.

Security Relevance

A burst of SYN packets directed at many destination ports on one host may indicate port scanning. However, this pattern alone does not prove malicious activity; legitimate inventory tools and authorized security assessments can generate similar traffic.

A security analyst can correlate the source IP, destination ports, packet timing, TCP flags, and target responses to investigate the activity.

6. Evidence and Screenshots

The following evidence should be included in the Findings folder:

i. **Host discovery:** Terminal showing the -sn scan.

ii. **SYN scan:** Terminal showing the -sS results.

iii. **Service detection:** Terminal showing the -sV output.

iv. **OS detection:** Terminal showing the -O output.

v. **Aggressive detection:** Terminal showing the -A output.

vi. **Wireshark SYN traffic:** Packets selected with the SYN display filter.

vii. **Saved files:** The Nmap output directory containing .txt and .xml files and the saved packet capture.

Screenshots should show the command, target, and relevant results wherever practical.

7. Conclusion

This practical demonstrated how Nmap can be used for host discovery, port scanning, service identification, operating system estimation, and broader network reconnaissance in an authorized lab environment.

Wireshark provided packet-level visibility into the TCP SYN scan, helping connect Nmap's scan results with the network traffic that a security monitoring system may observe.

The exercise established a foundation for identifying scanning behavior and understanding how network traffic can be investigated in a Security Operations Center (SOC).